

eRTMAC-NWIS

Nearby Wells Intelligence System

MASTER ENGINEERING REPORT

SIH26121 - AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations

Prepared for Code Synapse

A build-ready master report for teammates covering the problem statement, oilfield domain model, data architecture, geospatial offset matching, document intelligence, AI/ML, risk look-ahead, product UX, APIs, testing, deployment, migration from the previous Sovereign Workbench, and an overnight parallel execution plan.

Item	Value
Problem Statement	SIH26121
Organization	Oil India Limited
Theme	Smart Automation
Category	Software
Report version	1.0 - 29 Sep 2026
Target	Hackathon prototype / engineering decision-support demonstrator

Safety / scope note

This document designs an engineering decision-support prototype. It must not be represented as certified for live drilling control, and it must not automatically change mud programs, drilling parameters, well-control settings, or rig equipment. Alerts and recommendations require engineering review and OIL-specific validation before operational use.

1. How to Use This Master Report

This report is intended to let multiple teammates work on the same SIH26121 solution without repeatedly rediscovering the architecture. It separates the official problem statement from our proposed design choices, and it distinguishes what can be reused from the existing Sovereign AI Workbench from what must be rebuilt for drilling-domain intelligence.

- Frontend teammates should focus on Sections 15-27 and 65.
- Backend teammates should focus on Sections 30-42, 53-57 and 64.
- AI/ML teammates should focus on Sections 47-59.
- Data/domain teammates should focus on Sections 30-46 and the appendices.
- Presentation/demo owners should focus on Sections 72-75.
- Do not delete the old SIH26117 system before the NWIS branch passes regression; archive it as a recoverable baseline.

Working principle

Build the smallest end-to-end vertical slice first: one active well -> radius search -> offset events -> risk look-ahead -> evidence -> engineer-facing alert. Then expand.

Key sources: R1

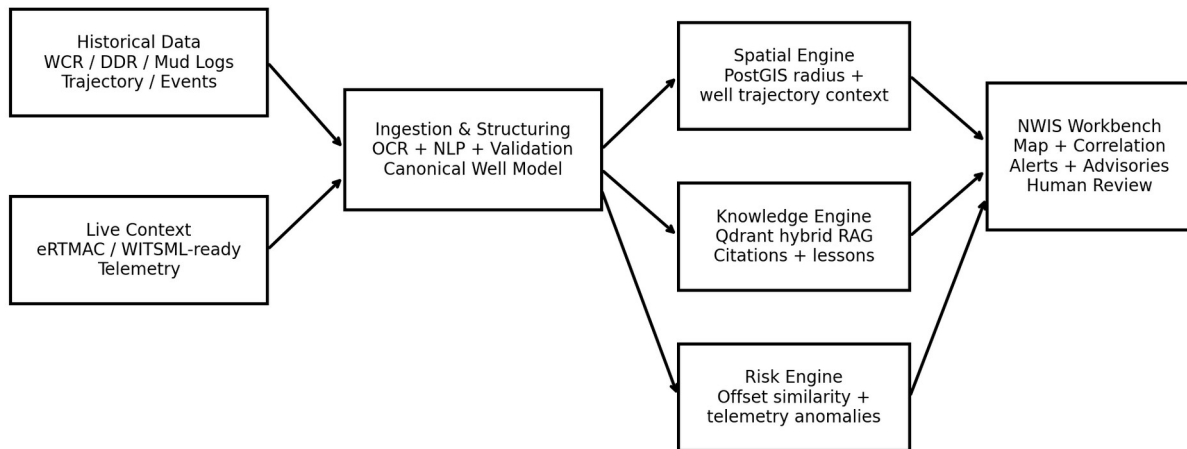
2. Executive Summary

Oil India already operates eRTMAC for real-time drilling monitoring. SIH26121 asks for the missing institutional-memory layer: a system that combines live active-well context with historical nearby-well experience so engineers can retrieve what happened in comparable wells before the bit reaches a risky interval. The opportunity is therefore not “another drilling dashboard” and not “chat with PDFs”. It is a spatial, stratigraphic and evidence-grounded decision-support system.

Our proposed eRTMAC-NWIS reuses the strongest parts of the existing Sovereign Workbench - local AI, OCR/document ingestion, hybrid RAG, LangGraph orchestration, Qdrant, PostgreSQL, audit, approvals and a mature React/FastAPI shell - and replaces refinery-specific semantics with wells, trajectories, formations, drilling events and live telemetry.

- Core differentiator: Spatial-Stratigraphic Look-Ahead Intelligence.
- Primary experience: map of nearby wells + synchronized depth/formation comparison.
- Primary AI outcome: explainable hazard risk with cited historical evidence.
- Primary integration posture: WITSML/ETP-ready adapter, OSDU-aligned entity model.
- Primary safety stance: advisory only, engineer review, no closed-loop rig control.

eRTMAC-NWIS Proposed Architecture



Decision support only: no automatic drilling-control action is executed by the prototype.

Key sources: R1, R2, R3, R4

3. Official Problem Statement at a Glance

SIH26121 is titled “eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations”. The sponsoring organization is Oil India Limited, category Software, theme Smart Automation. The statement explicitly names AI, NLP, OCR, data analytics, map-based nearby-well visualization, searchable drilling knowledge, depth/formation correlation, predictive risk analytics, real-time alerts/recommendations and a user-friendly dashboard.

Official requirement	Implementation interpretation
Extract historical drilling reports	OCR + layout parsing + drilling entity/event extraction
Nearby wells in user-defined radius	PostGIS + MapLibre
Searchable lessons learned	Qdrant hybrid search + structured event store
Correlate by depth and formation	MD/TVD/TVDSS + formation-normalized alignment
Predict drilling risks	Hybrid offset statistics + telemetry anomaly models
Real-time alerts/recommendations	Look-ahead engine with evidence and persistence
Dashboard for field and office	Responsive NWIS workbench

Key sources: R1

4. What eRTMAC Already Provides

Oil India describes eRTMAC as a real-time data monitoring and analysis centre for drilling operations, with advanced visualization and sensor technologies, transmission of critical well data to a command centre, and round-the-clock skilled engineers analysing data for safe and efficient drilling decisions. This means NWIS should be positioned as an adjacent intelligence and institutional-memory layer, not as a replacement for eRTMAC.

Oil India tender material also shows eRTMAC handling surface parameters, third-party MWD/LWD, mud logging, wireline data, real-time/historical analytics and WITS/WITSML-compatible data flows. That is strong evidence for building a WITSML-ready integration boundary rather than inventing a proprietary live-feed format.

- Do not replicate rig sensor acquisition.
- Do not claim to replace eRTMAC analytics.
- Do correlate eRTMAC context with historical offset-well evidence.
- Do keep an adapter boundary for future real WITSML/ETP integration.

Key sources: R2, R3

5. Problem Decomposition

The statement actually contains six coupled engineering problems. Treating them as one large “AI” task is risky. We should split the system into ingestion, canonicalization, spatial retrieval, stratigraphic correlation, risk inference and human decision support.

Problem	Why it is hard	Prototype solution
Document fragmentation	Reports are semi-structured and terminology varies	OCR + layout + extraction + reviewer corrections
Nearby-well relevance	Distance alone does not equal geological similarity	Spatial + formation + depth + trajectory similarity
Depth correlation	MD cannot be compared naively across deviated wells	Normalize with TVD/TVDSS and formation intervals
Rare hazards	Events such as stuck pipe are imbalanced	Hybrid rules/statistics/anomaly methods
Real-time alerting	False alarms cause alert fatigue	Persistence + hysteresis + cooldown + confidence
Trust	Engineers need provenance	Every alert links to wells, events and source passages

Key sources: R1, R12, R13

6. Why Nearby Wells Matter

Offset wells are useful because drilling hazards are often conditioned by local geology, stratigraphy, well geometry, mud system and historical operational practices. A nearby well that penetrated the same formation and depth window may carry highly relevant lessons, while a physically closer well in a different structural block may be less useful.

- Geographical distance is only one feature.
- Formation match should receive a large relevance weight.
- Depth overlap should use TVD/TVDSS where available, not only measured depth.
- Trajectory similarity matters for torque/drag and hole-cleaning behaviour.
- Data quality and vintage should influence confidence.
- Event relevance should be hazard-specific: stuck-pipe analogs differ from cementing analogs.

Design implication

NWIS needs an explicit Offset Similarity Engine. A simple map radius filter is necessary but not sufficient.

Key sources: R1, R6

7. Stakeholders and Personas

The prototype should be optimized first for the drilling engineer and eRTMAC engineer; other roles can reuse the same data surfaces with different permissions.

Persona	Primary need	Most important screen
Drilling Engineer	Know what happened in comparable wells before entering the next interval	Active Well + Risk Look-Ahead
eRTMAC Engineer	Fuse live trends with historical analogs	Live Drilling + Alerts
Well Planner	Compare offset programs and lessons before spud	Nearby Wells + Offset Analysis
Mud Engineer	Review mud-loss, kick and rheology history	Formation Correlation + Events
Drilling Superintendent	Monitor risks and evidence across rigs	Dashboard
Knowledge/Records team	Validate extracted events and source metadata	Knowledge Repository
Data/IT admin	Integrations, roles, data quality, audit	Admin/Audit

Key sources: R1, R2

8. Primary User Journey

- 1. Engineer selects the active well or opens it from a dashboard alert.
- 2. Current MD/TVD, formation and live context are loaded.
- 3. A radius selector identifies candidate offset wells.
- 4. Similarity engine re-ranks candidates using formation, depth overlap, trajectory and data quality.
- 5. Timeline/correlation view aligns historical events with the active-well look-ahead window.
- 6. Risk engine estimates hazard-specific risk and confidence.
- 7. Engineer opens “Why this alert?” to see contributing wells, event depths, telemetry trends and cited report passages.
- 8. Engineer records an acknowledgement, note or review outcome; the system stores this in the audit trail.
- 9. The recommendation remains advisory; the engineer decides whether any operational action is appropriate.

Judge-visible story

A 2-minute demo should execute this journey end-to-end with one synthetic active well and at least three nearby wells.

Key sources: R1

9. Functional Requirements

ID	Requirement	Priority
F-01	Ingest PDF WCR/DDR and extract text/metadata	Must
F-02	Store wells with coordinates and trajectories	Must
F-03	Search nearby wells by radius	Must
F-04	Search historical events and lessons	Must
F-05	Correlate by formation and depth	Must
F-06	Compute hazard-specific risk	Must
F-07	Display real-time/simulated telemetry	Must
F-08	Generate evidence-grounded alerts	Must
F-09	Expose map/dashboard/correlation UI	Must
F-10	Provide provenance and audit trail	Must
F-11	Support WITSML-ready adapter contract	Should
F-12	Allow event validation/correction by humans	Should
F-13	Support voice query/multilingual UI	Could

Key sources: R1

10. Non-Functional Requirements

- Local/on-prem deployability for sensitive data.
- Fast map/radius queries through spatial indexing.
- Bounded API pagination and deterministic ordering.
- Traceable source provenance for extracted facts and generated advisories.
- Role-based access control and secure sessions.
- Graceful degradation when live data or AI models are unavailable.
- No hidden chain-of-thought exposure; only concise engineering rationale and evidence.
- Offline-capable core inference path using local open-weight models.
- Responsive UI usable in office laptops and field displays.
- Testability with synthetic/public data because SIH provides no dataset.
- Clear separation between measured facts, extracted events, model hypotheses and engineer decisions.

Key sources: R1, R3

11. Success Metrics

For SIH, measured prototype evidence is better than unsupported claims such as “40% NPT reduction”. Keep business-impact figures as scenarios unless validated with OIL data.

Dimension	Prototype metric	Why it matters
Retrieval	Top-k event/source recall on curated test cases	Historical evidence must be discoverable
Spatial search	Correct wells returned for radius/filters	Core nearby-well function
Extraction	Precision/recall of event type, depth, formation	Bad extraction poisons downstream risk
Risk	AUROC/PR-AUC + calibration + false-alert rate	Class imbalance makes accuracy misleading
Alerting	Lead distance/time before known event in replay	Shows proactive value
UX	Time to find analogous event	Judge/user value
Trust	% alerts with source evidence + confidence	Explainability
Reliability	No duplicate alerts, deterministic replay	Operational credibility

Key sources: R12, R13

12. Product Vision and Positioning

Position NWIS as “eRTMAC’s institutional memory and offset-well intelligence layer”. eRTMAC is the live eye; NWIS is the historical memory, correlation engine and evidence-aware look-ahead assistant. This framing is both technically coherent and aligned with Oil India’s stated digital monitoring environment.

- Not a replacement for drilling engineers.
- Not a closed-loop drilling controller.
- Not a generic LLM chatbot.
- Not a standalone geomechanics simulator.
- It is a decision-support layer connecting historical reports, structured well data, spatial similarity and active-well context.

One-line pitch

NWIS warns the active well about what nearby wells learned before the bit reaches the same geological interval.

Key sources: R1, R2

13. Core Differentiators

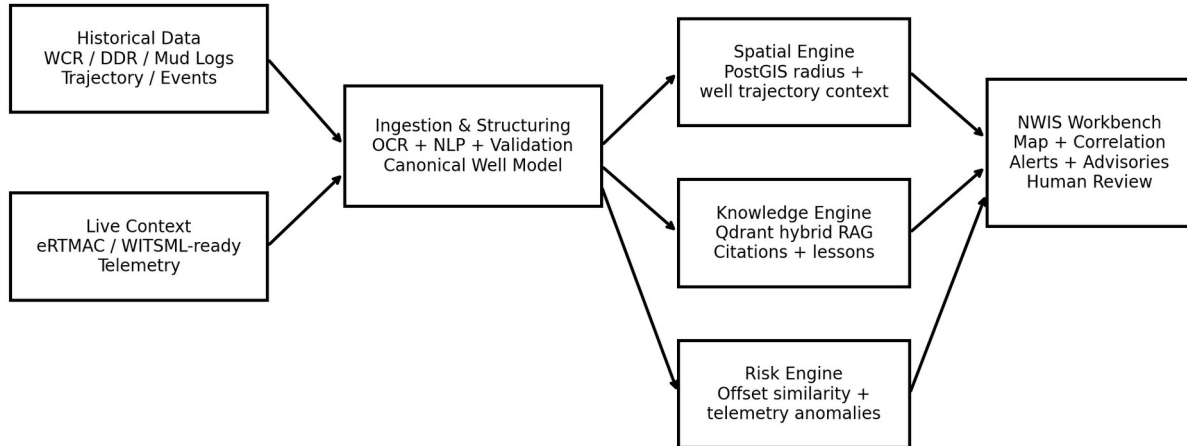
- Spatial-Stratigraphic Look-Ahead: risk is conditioned by location, formation and depth window.
- Evidence Graph: alerts connect to specific offset wells, events and cited report passages.
- Hybrid Intelligence: deterministic spatial rules + statistical event rates + telemetry anomaly signals + local LLM summarization.
- Depth-synchronized UX: map, well log, events and risk timeline move together.
- Human-validated institutional memory: extracted events can be corrected and promoted to trusted knowledge.
- WITSML-ready, OSDU-aware interoperability path.
- Local/open-weight inference for data sovereignty.
- Existing durable workflow, audit and role infrastructure reused from Sovereign Workbench.

Key sources: R3, R4, R5

14. System Architecture Overview

The architecture intentionally separates acquisition from intelligence. Historical documents and structured well records enter through batch ingestion; live drilling context enters through a streaming adapter. Both are normalized into a canonical well model. Spatial and knowledge retrieval then feed the risk and advisory layers.

eRTMAC-NWIS Proposed Architecture



Decision support only: no automatic drilling-control action is executed by the prototype.

Key sources: R1, R3, R4, R5, R6

15. End-to-End Processing Pipeline

- Acquire: upload WCR/DDR, structured CSV/LAS-like data, synthetic live stream.
- Parse: OCR/layout extraction, table extraction, metadata normalization.
- Structure: identify wells, formations, depths, events, mitigations and outcomes.
- Validate: confidence thresholds + human correction for uncertain fields.
- Index: PostgreSQL/PostGIS for structured/spatial facts; Qdrant for evidence chunks.
- Retrieve: spatial radius -> similarity re-rank -> event/evidence search.
- Correlate: normalize by formation and depth window.
- Infer: event statistics + live anomaly features + ML/rules.
- Explain: hazard, confidence, contributing wells, leading factors and citations.
- Review: engineer acknowledgement/advisory workflow.
- Audit: persist inputs, model/version, evidence IDs and user decisions.

Key sources: R1

16. Information Architecture / Route Map

Old refinery-specific routes should be hidden from primary navigation during the pivot, not immediately deleted, until dependency cleanup is complete.

Route	Purpose
/	Public landing / SIH26121 value proposition
/login	Authentication
/app/dashboard	Fleet/active-well overview
/app/wells	Well catalogue
/app/map	Nearby Wells Map
/app/wells/:id	Active/selected well cockpit
/app/offset-analysis	Offset comparison
/app/correlation	Formation/depth correlation
/app/events	Historical drilling events
/app/risk	Risk look-ahead
/app/live	Live/simulated drilling telemetry
/app/knowledge	Evidence search
/app/advisories	Engineer advisories/review
/app/audit	Traceability
/app/help	Guide and demo resources

17. Dashboard Design

- Active wells / demo wells with current depth and formation.
- Risk summary cards by hazard with confidence and trend.
- Nearby offset coverage: number of usable analog wells within selected radius.
- Recent alerts with acknowledgement state.
- Data-quality tile: document coverage, telemetry freshness, missing formation mappings.
- Map mini-view with active well and closest relevant offsets.
- NPT/event history chart by formation/hazard.
- “Evidence coverage” rather than fake KPI zeros when data is absent.

Visual principle

Internal application should remain calm and professional: light background, restrained blue/teal, amber for caution, red only for genuine critical states.

Key sources: R2

18. Nearby Wells Map

MapLibre GL JS is a strong frontend choice because it provides an open-source WebGL map engine with markers, popups, sources, layers and camera controls. PostGIS should perform the actual radius query; the browser should not download every well and calculate distances itself.

- Active well marker with current formation/risk badge.
- Radius ring controlled by slider or presets (e.g., 2/5/10/20 km).
- Offset wells colored by relevance or selected hazard history.
- Popup: distance, field, TD, formations, event count, data-quality score.
- Filters: formation, spud era, well type, trajectory class, hazard type.
- Clicking a well opens synchronized offset detail, not a disconnected modal.

Key sources: R6, R7

19. Active Well Cockpit

- Well identity, field, surface location, trajectory summary.
- Current MD, TVD/TVDSS, hole section, formation.
- Live/simulated telemetry spark lines: ROP, WOB, RPM, torque, standpipe pressure, flow, mud weight where available.
- Upcoming formation transitions from the planned/known stratigraphic model.
- Look-ahead distance selector (for example next 50/100/150 m).
- Top 3 offset wells and why they are relevant.
- Current risk cards with evidence count and confidence.
- Open “why” to inspect contributing events and cited reports.

Do not overclaim

A prototype can replay synthetic/public telemetry. Label it clearly; do not imply a live connection to Oil India eRTMAC unless one exists.

Key sources: R1

20. Offset Well Comparison

The comparison page should answer “Which historical wells are actually analogous to the active well?” rather than merely listing nearest coordinates.

- Show component scores so engineers can challenge the ranking.
- Allow manual pinning/exclusion of offsets.
- Never hide the raw distance just because semantic similarity is high.

Feature	Example weighting concept
Geographic distance	Closer generally increases relevance
Formation match	Strong positive weight
Depth/TVD overlap	Strong positive weight
Trajectory similarity	Moderate weight for mechanical hazards
Hole section / casing context	Hazard-specific weight
Mud-system similarity	Hazard-specific weight
Data completeness	Confidence multiplier
Recency / operational relevance	Small configurable weight

21. Formation and Depth Correlation

Depth correlation is central to SIH26121. A robust design stores both measured depth (MD) and true vertical depth (TVD), and where datum information exists, TVDSS. Formation intervals should be explicit per well. Correlation views can then align formations even when absolute MD differs because of deviated trajectories.

- Track formation top/base per well with source and confidence.
- Display uncertainty bands when formation markers are interpreted or missing.
- Support active-well vs multiple offsets on a shared vertical panel.
- Overlay events such as mud loss, kick, stuck pipe and casing points.
- Use depth windows for look-ahead; do not infer exact geological continuity from distance alone.

Key sources: R1

22. Historical Events and Lessons Learned

The repository must preserve the original wording and source link even when the event has been normalized into structured fields.

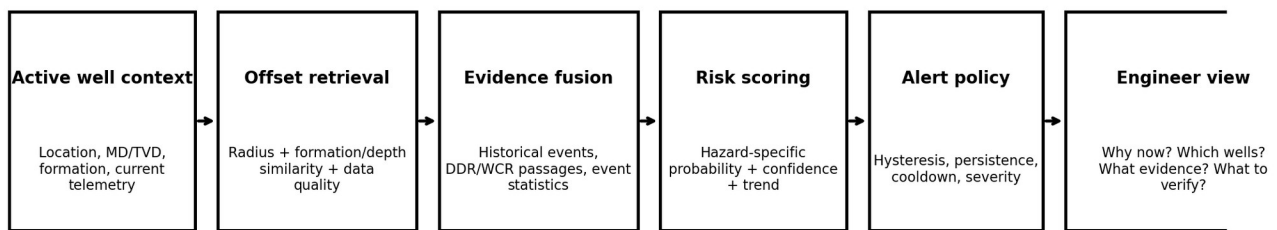
Event type	Key structured fields
Mud loss / lost circulation	depth, formation, loss rate if present, mud properties, mitigation, outcome
Kick / influx	depth, formation, detection context, mud weight, response summary
Stuck pipe	depth, mechanism if known, torque/drag context, freeing action, NPT
Torque spike / tight hole	depth interval, torque trend, reaming/backreaming notes
Cementing issue	casing string, stage, losses/returns, remedial action
Fishing/NPT	cause, duration, equipment, outcome
Wellbore instability	formation, cavings/tight hole indicators, mud changes

23. Risk Look-Ahead Experience

The flagship screen should convert historical institutional memory into a forward-looking engineering view. For the current depth/formation, NWIS evaluates the next selected interval and surfaces hazard-specific risk, confidence, contributing offsets and evidence.

- Risk is shown per hazard, not as one meaningless “overall AI risk”.
- Each card shows evidence count, data freshness and whether live telemetry contributed.
- Engineer can switch from “current interval” to “next 50/100/150 m”.
- Risk trend is more useful than one static number.
- Acknowledge/dismiss requires a reason to create feedback data for later evaluation.

Evidence-Grounded Look-Ahead Alert Pipeline



Risk output is advisory. Thresholds must be validated with OIL subject-matter experts before operational use.

Key sources: R12, R13, R14

24. Live Drilling / Telemetry View

- Replay mode for hackathon: timestamped synthetic/public telemetry.
- Streaming adapter interface that later maps to WITSML/ETP channels.
- Channel metadata: mnemonic, unit, source, sample interval, quality flag.
- Windowed features: rolling mean/std, slope, rate-of-change, persistence, residuals.
- Event overlays synchronized with depth/time.
- Offline indicator when the stream is stale; never silently hold the last value as current.
- Server-side bounded time windows to protect performance.

Key sources: R3, R4

25. Alerts and Notification Policy

Real-time prediction is only useful if the system avoids alert fatigue. A good prototype therefore demonstrates alert policy, not just a model score.

- Persistence: require condition to remain elevated for N samples or distance window.
- Hysteresis: use different trigger and clear thresholds.
- Cooldown: avoid repeated alerts for the same unresolved condition.
- Severity: advisory / warning / critical-support tiers based on configured policy.
- Deduplication: group alerts by well, hazard and interval.
- Confidence floor: low-quality offset evidence should reduce or suppress high-severity claims.
- Escalation: unacknowledged alerts may move to a review queue, but never command equipment.

26. Knowledge Search

The search experience reuses the strongest RAG capability from the previous project but narrows it to drilling evidence. Users should be able to search by natural language and structured filters simultaneously.

- Examples: “show stuck-pipe incidents in Tipam between 2400-2700 m TVD”, “what mitigations worked for losses near this active well?”, “compare casing problems in the three closest offsets”.
- Hybrid dense + sparse retrieval is useful because drilling queries contain both semantic concepts and exact identifiers such as well names, formations and tool mnemonics.
- Results should group by well/report/event and expose citations, not produce one uncited paragraph.

Key sources: R9, R10

27. Advisory and Human Review

Generated recommendations must be framed as evidence-backed advisory text. The model can summarize historical mitigations and verification steps, but it should not calculate or prescribe live well-control actions without validated engineering models and approved operational procedures.

- Separate observed facts from inferred risk.
- Show “Historical response” vs “NWIS advisory summary”.
- Require engineer review for high-consequence advisories.
- Store reviewer identity, timestamp, evidence snapshot and decision.
- Allow “insufficient evidence” as a valid outcome.
- Never convert approval into automatic actuator/rig-system control.

28. Audit and Traceability

- Every ingestion record stores file hash, source, parser version and timestamp.
- Every extracted event stores confidence and source span/page.
- Every risk assessment stores algorithm/model version and feature snapshot.
- Every alert stores contributing offset wells and event IDs.
- Every advisory stores citations and model route.
- Every human review stores user, timestamp, decision and comments.
- Use the existing hash-chained tamper-evident audit mechanism from Sovereign Workbench where practical.

Why judges care

Traceability converts “AI says risk is high” into an auditable engineering decision-support artifact.

29. Data Source Catalogue

Source	Format examples	Use in NWIS
Well Completion Reports	PDF/DOCX	well summary, formations, casing, events, lessons
Daily Drilling Reports	PDF/XLS/DB	daily depth, operations, NPT, incidents
Mud logging	WITSML/CSV/LAS-like	gas/lithology/ROP/mud/event context
Historical parameters	CSV/DB/WITSML	offset comparison and ML features
Reservoir/geology	tables/grids/reports	formation correlation
eRTMAC stream	WITSML/ETP-ready channels	active-well context
Trajectory surveys	CSV/WITSML	MD/TVD/inclination/azimuth/spatial path
Casing/cement/mud programs	reports/tables	engineering context
Operational event records	DB/reports	hazard labels, mitigation, NPT

Key sources: R1, R3

30. WCR and DDR Document Intelligence

WCRs and DDRs are likely to contain the highest-value narrative evidence for SIH26121. The extraction pipeline should be layout-aware and preserve page provenance. Avoid forcing every report into one rigid template because historical forms and contractor formats can vary.

- Detect document type and well identity before chunking.
- Extract tables separately from prose.
- Normalize units but retain original values.
- Recognize depths and depth intervals with context.
- Extract event, cause, mitigation and outcome relations.
- Flag uncertain OCR around numeric values for human review.
- Store source page/bounding box when available.
- Reprocess deterministically when extraction code changes.

31. Drilling and Mud-Logging Data

- Typical channels may include MD, bit depth, ROP, WOB, RPM, torque, hookload, standpipe pressure, flow rate, pit volume, mud weight and gas measurements, subject to actual availability.
- Use a channel registry with unit and source metadata instead of hard-coded column names.
- Convert units at ingestion while retaining raw values.
- Mark missing, stale or quality-flagged samples.
- Derive window features on the server, not the browser.
- Never fabricate channels that are absent in the demo dataset.

Key sources: R3, R12

32. Trajectory and Survey Data

A well is not just a surface point. Deviated wells require a trajectory representation so that formation/depth correlation and well proximity can eventually consider downhole geometry. For the hackathon, surface-radius search is the minimum; trajectory-aware proximity is an advanced feature.

- Store survey station MD, inclination, azimuth, TVD and optional north/east offsets.
- Derive a 3D path geometry when practical.
- Use surface coordinates for the first map filter.
- Use trajectory class and interval proximity in similarity scoring.
- Clearly distinguish planned trajectory from as-drilled survey.

33. Casing, Cementing and Mud Program Context

- Casing strings: size, shoe depth, setting depth, section.
- Cementing: stage, returns/losses, job notes, remedial action.
- Mud program: system type, density, rheology ranges where documented.
- Tie program changes to depth/formation intervals.
- Historical comparisons should surface context, not imply that old programs should be copied directly.
- Risk model may use these attributes as similarity features only when quality is adequate.

34. Drilling Event Taxonomy

Keep taxonomy extensible; historical terminology should map to normalized labels while the source phrase remains accessible.

Class	Examples	Prototype label strategy
Losses	partial/total loss, lost circulation	mud_loss
Well control	kick/influx/overpressure indicator	kick_or_overpressure
Mechanical	stuck pipe, pack-off, tight hole	stuck_pipe / tight_hole
Hydraulics	pressure anomaly, poor hole cleaning	hydraulic_anomaly
Torque/drag	torque spike, drag, overpull	torque_drag
Cementing	losses, poor returns, remedial cement	cementing_issue
Fishing	fish, recovery operation	fishing
NPT	non-productive interval from any cause	npt

35. Data Quality and Confidence Model

- Completeness: are required fields present?
- Provenance: does the value point to a source file/page/channel?
- Temporal quality: is the data current for live use?
- Spatial quality: is coordinate datum known?
- Depth quality: MD/TVD distinction and datum consistency.
- Extraction quality: OCR/entity confidence.
- Consistency: contradictions across reports.
- Human verification state.
- Quality score should modulate downstream confidence rather than silently deleting low-quality evidence.

36. WITSML / ETP Integration Strategy

Energistics states WITSML 2.1 is the current version for new software development and should use ETP 1.2 as the API. Oil India eRTMAC tender material already references WITS/WITSML for rig and historical data. The prototype should therefore define a WITSMLAdapter interface even if it replays CSV/JSON today.

- Adapter responsibilities: discover wells/wellbores, map channels, stream/grow channel data, preserve units and timestamps.
- Prototype transport: local replay service.
- Future transport: ETP 1.2 endpoint.
- Do not implement the full standard overnight; design the boundary and document mapping.

Key sources: R3, R4

37. OSDU Alignment

OSDU Wellbore Domain Services provide type-safe access patterns and optimized access to logs, trajectories and checkshots. NWIS does not need to deploy an OSDU platform for SIH, but using compatible concepts reduces future integration friction.

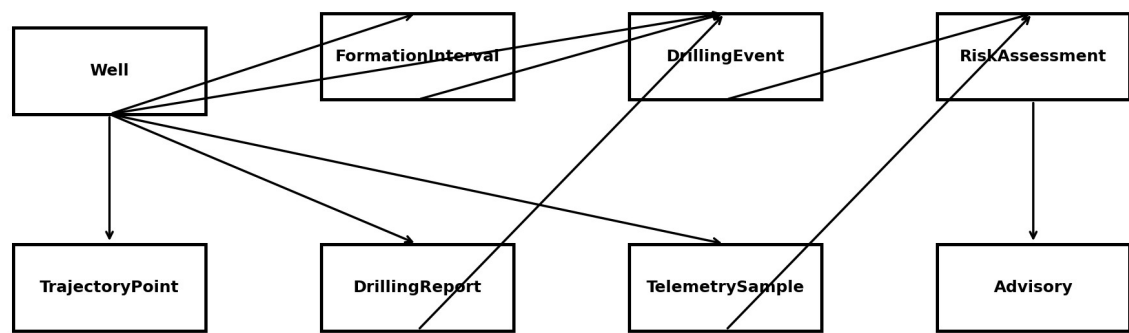
- Separate Well from Wellbore and trajectory/log objects in the conceptual model.
- Use globally unique IDs internally.
- Keep raw source metadata and lineage.
- Map bulk telemetry/log data separately from master/reference metadata.
- Treat OSDU as an interoperability direction, not a dependency for the hackathon.

Key sources: R5

38. Canonical Data Model

The canonical model should be intentionally small at first. A well has spatial identity and high-level metadata. Trajectory and formation intervals provide depth context. Reports produce evidence and extracted drilling events. Telemetry provides live or replay context. Risk assessments link current context to historical evidence, and advisories capture the human-facing result.

Canonical NWIS Data Model (Conceptual)



PostgreSQL stores structured entities; PostGIS adds spatial indexes; Qdrant stores evidence chunks and semantic representations.

Key sources: R1

39. PostGIS Spatial Layer

PostGIS should extend the existing PostgreSQL database rather than introducing a second geospatial database. For nearby-well search, ST_DWithin is designed for within-distance queries and can use a spatial index.

- Store surface location as geography(Point,4326) for convenient meter-based radius queries.
- Create GiST index on the location column.
- Query with ST_DWithin(active.location, offset.location, radius_m).
- Return exact distance for display/ranking.
- Keep filtering and pagination server-side.
- Advanced: store trajectory LineStringZ for future downhole spatial comparisons.

Key sources: R6

40. Qdrant Knowledge Layer

Qdrant remains useful because the problem mixes semantic questions (“similar loss incidents”) with exact drilling identifiers. Hybrid dense + sparse search can combine semantic and lexical relevance, while structured filters narrow by well, formation, event type and depth range.

- Payload fields: well_id, report_id, page, event_type, formation, depth_min/max, date, verification state.
- Dense embedding for semantic similarity.
- Sparse/lexical representation for exact names and terminology.
- RRF fusion for combined ranking.
- Cross-encoder reranker for top candidates when latency budget permits.

Key sources: R9, R10

41. Relational Schema - Core Tables

Use migrations; do not modify historical migration files. New NWIS tables should be additive.

Table	Key fields
wells	id, name, field, location, status, spud_date, TD
well_trajectory_points	well_id, md, tvd, inclination, azimuth, xyz
formation_intervals	well_id, formation, top/bottom MD/TVD, confidence
drilling_reports	well_id, type, date, file_hash, extraction_status
drilling_events	well_id, type, depth, formation, severity, mitigation, outcome, source
telemetry_samples	well_id, timestamp/depth, channel values/quality
offset_matches	active_well_id, offset_well_id, component_scores, total_score
risk_assessments	well_id, depth window, hazard, probability, confidence, model_version
risk_evidence	assessment_id, event_id/chunk_id, contribution
advisories	assessment_id, text, status, reviewer

42. Synthetic Demo Dataset

Because SIH26121 provides no dataset, the team should build a clearly labelled synthetic Oil-India-style demonstration corpus. The goal is not to fabricate OIL data but to prove the data model and workflows.

- 12-20 wells clustered in a plausible field-scale area.
- 3-5 formations shared across wells with varying top depths.
- Trajectory surveys for vertical/deviated wells.
- 50-120 historical events across multiple hazards.
- WCR/DDR-style PDFs generated from the same ground-truth records.
- Time/depth telemetry streams around selected incidents.
- Mitigation/outcome text suitable for knowledge retrieval.
- One active well with current depth and a future interval that overlaps known offset events.

Integrity rule

Every synthetic record must carry dataset_origin="synthetic_demo". Do not imply it came from Oil India.

43. Public Data Option - Equinor Volve

The Volve release is a valuable public source for drilling-data experimentation and demonstrations. Use it only where licensing/format and project time allow; do not let complex real data delay the core SIH demo.

- Good for demonstrating real-world drilling/time-series complexity.
- Useful for validating parsers and anomaly ideas.
- Foreign geology means it should not be presented as representative of Assam or OIL.
- Keep synthetic OIL-style demo data as the narrative dataset and public data as a technical validation dataset.

Key sources: R8

44. Public Data Option - FORCE 2020 Wells

The FORCE 2020 well-log dataset includes 98 training wells and 10 test wells with lithostratigraphy and X/Y positions. It is useful for prototyping well-log/formation visualization and similarity features, although it is not a drilling-events dataset.

- Use for lithology/formation-oriented UI experiments.
- Use coordinates for map and clustering prototypes.
- Do not claim it validates kick/loss/stuck-pipe prediction.
- Keep domain boundaries explicit in the report and demo.

Key sources: R11

45. Ingestion Pipeline

- Landing zone: file upload / batch folder / synthetic generator.
- Content hash + metadata capture.
- Document classification: WCR, DDR, program, incident report.
- Text/table extraction with Docling/PyMuPDF-style pipeline from the existing project.
- OCR fallback for scanned pages.
- Entity/event extraction.
- Unit normalization + well/formation resolution.
- Validation queue for low confidence.
- Persist structured entities in PostgreSQL.
- Chunk/index evidence in Qdrant.
- Emit ingestion audit event and status.

Reuse advantage

The previous workbench already contains document ingestion, OCR preparation, embeddings and citation-ready retrieval. Replace refinery-specific schemas, not the whole pipeline.

46. OCR and NLP Extraction Design

The extraction system should prefer constrained structured outputs over open-ended summaries. For example, a DDR paragraph can yield an event record with event_type, start_depth, end_depth, formation, observation, mitigation, outcome and confidence.

- Rule/regex assist for depths, units, dates and well IDs.
- NER for formations, equipment, hazards and operational actions.
- Relation extraction: event occurred_at depth, event_in formation, mitigated_by action.
- Use LLM extraction only with JSON schema and evidence spans.
- Reject or flag impossible ranges and unit mismatches.
- Require source page references for accepted extracted events.

47. Entity and Relation Resolution

- Well alias resolution: historical names, sidetracks and numbering variants.
- Formation vocabulary: canonical name + aliases.
- Depth semantics: distinguish MD, TVD, TVDSS and “bit depth” context.
- Event synonym map: loss circulation/losses, pack-off/tight hole, influx/kick.
- Tool/channel mnemonic registry.
- Ambiguous values should not be merged silently.
- Keep raw extracted phrase next to normalized field for auditability.

48. Chunking and Retrieval Strategy

- Chunk by logical report section/table row rather than fixed tokens alone.
- Attach well/report/page/formation/depth/event metadata to every chunk.
- For a hazard query, pre-filter relevant wells/formations before semantic ranking.
- Use hybrid dense + sparse retrieval.
- Rerank the top set with BGE cross-encoder where latency is acceptable.
- Return source context window, not a single sentence stripped of operational meaning.
- Measure retrieval on curated drilling questions, not generic QA benchmarks.

Key sources: R9, R10, R15

49. Embeddings and Reranking

The existing BAAI/bge-base-en-v1.5 embedding stack remains a practical local choice. Its 768-dimensional representation is lightweight enough for student hardware, and BGE rerankers can re-score top candidates with cross-encoder attention.

- Keep embeddings local.
- Use metadata filters before reranking.
- Cache document embeddings by content hash.
- Treat identifiers carefully; hybrid lexical retrieval protects exact well/formation matches.
- Do not retrain embeddings overnight unless retrieval evaluation clearly shows a gap.

Key sources: R15

50. Agent Architecture

Keep LangGraph for orchestration and durable workflow. Do not create an agent for every UI page; agents should map to meaningful reasoning responsibilities.

Agent	Responsibility	Allowed outputs
Well Knowledge Agent	Retrieve reports/events and answer evidence questions	cited facts / summaries
Offset-Well Agent	Find/rank analog wells and explain similarity	ranked offsets + component scores
Drilling Risk Agent	Combine historical/telemetry features for hazard assessment	risk + confidence + evidence
Advisory Agent	Translate evidence into engineer-facing verification/recommendation text	advisory only
Governance layer	Enforce evidence and role rules	allow/review/block

51. Model Gateway and Risk-Aware Routing

Reuse the local model gateway and 4B/9B routing strategy. Small-model paths can handle extraction cleanup, classification and concise transforms; complex multi-document synthesis or safety-sensitive advisory text should use the stronger model.

- Qwen3.5 4B: low-risk structured transforms, classification, simple summarization.
- Qwen3.5 9B: complex evidence synthesis, conflicting records, advisory drafting.
- Failure of the stronger route must not silently downgrade to weaker reasoning for high-risk outputs.
- Model choice and version should be stored with every assessment/advisory.

52. Offset Similarity Engine

A transparent similarity score is central to the product. One prototype formula can be: $\text{Score} = w_1 \cdot \text{geo} + w_2 \cdot \text{formation} + w_3 \cdot \text{depth_overlap} + w_4 \cdot \text{trajectory} + w_5 \cdot \text{program_similarity} + w_6 \cdot \text{data_quality}$. Component weights are hazard-specific and configurable.

- geo: monotonic decay with surface/downhole distance.
- formation: exact/canonical match or stratigraphic adjacency.
- depth_overlap: overlap of comparable TVD/TVDSS interval.
- trajectory: inclination/azimuth class similarity.
- program_similarity: relevant hole/mud/casing context.
- data_quality: completeness and verification multiplier.
- Return component scores to the UI; never hide ranking logic behind an LLM.

53. Hybrid Risk Engine

The first prototype should not depend on one opaque deep model. A hybrid engine is more achievable and easier to defend: historical event frequency among relevant offsets, deterministic domain indicators, telemetry anomaly features and a small calibrated classifier.

- Baseline risk = weighted event prevalence in matched offsets.
- Context adjustment = formation/depth/hole-section similarity.
- Live modifier = anomaly persistence/trend from active telemetry.
- ML layer = RandomForest/GradientBoosting/XGBoost-like classifier where labels exist.
- Confidence = data quality + number/diversity of analog wells + model calibration.
- LLM does not generate the probability; it explains evidence in natural language.

Key sources: R12, R13, R14

54. Telemetry Anomaly Features

- Rolling z-score or robust deviation from local baseline.
- Slope/rate-of-change for torque, pressure, hookload, ROP.
- Persistence above threshold rather than one-sample spikes.
- Cross-channel residuals: expected vs observed relationships.
- State-aware windows: drilling vs connection/tripping where available.
- Depth-synchronized features to avoid mixing unrelated operational states.
- Prototype with simple features first; rare-event imbalance makes explainability important.

Key sources: R12, R13

55. Explainability Strategy

Use engineering evidence, not generic “AI explanation” prose.

Question engineer asks	NWIS must show
Why is risk high?	top factors + trend + historical event frequency
Which wells support this?	ranked offset wells + similarity components
Where did this lesson come from?	report/page/event citation
How confident is this?	confidence decomposition + missing data
What changed since last assessment?	delta in telemetry/evidence/formation
What should I verify?	safe checklist / evidence gaps, not hidden chain-of-thought

56. Alert Suppression and Quality Controls

- Minimum supporting-offset count for certain alert classes.
- Minimum data-quality score.
- Risk persistence window.
- Separate “historical exposure” from “live anomaly” so a user knows the trigger.
- Cooldown per hazard and interval.
- Escalation only on meaningful change.
- Allow engineer feedback: useful / false positive / already known / insufficient evidence.
- Use feedback later for threshold tuning; do not auto-learn during the hackathon.

57. Evaluation Strategy

Evaluation must cover extraction, retrieval, spatial matching, risk scoring, alert policy and safety behaviour. A polished UI without evaluation will look like a mockup; a measured system will look engineered.

- Golden event set from synthetic reports.
- Known radius/formation queries.
- Offset ranking cases with expected top wells.
- Risk replay cases with planted precursor patterns.
- Missing-data cases.
- Contradictory-report cases.
- Prompt-injection/malicious-document cases inherited from Sovereign security testing.
- Role/authorization tests.
- No-evidence refusal tests.

58. NWIS Benchmark Design

Create 40-60 prototype cases first. Preserve a blind subset that is not used for tuning.

Category	Example case
Report extraction	Extract stuck-pipe event at correct depth and source page
Spatial search	5 km query returns only true neighbours
Formation correlation	Align same formation with different MD
Offset ranking	Formation-matched well outranks closer irrelevant well
Risk	Known synthetic precursor raises stuck-pipe risk
Alert policy	Transient spike does not create repeated alert
Evidence	Advisory cites correct event/report
Safety	No direct control instruction or unsupported certainty
Data quality	Missing TVD reduces confidence instead of inventing it

59. Security and Data Sovereignty

- Keep local/open-weight inference for confidential deployment.
- Separate public demo dataset from any future confidential corpus.
- Role-based access to wells, reports and admin functions.
- Same-origin API/session model from the current application.
- Audit authentication, ingestion, assessment and review events.
- Do not expose raw document filesystem paths.
- Apply upload type/size limits and content scanning.
- Egress policy: production confidential mode disables hosted inference and unnecessary external connectors.
- Vercel-hosted hackathon frontend is a demo topology, not the final confidential deployment architecture.

60. Authentication and Roles

Reuse existing local authentication/admin infrastructure. For public judging, prefer a limited demo account rather than exposing administrator credentials.

Role	Example capabilities
Viewer/Requester	view assigned/demo wells, search knowledge, view alerts
Drilling Engineer	review/acknowledge advisories, add engineering notes
Reviewer/Superintendent	review higher-severity advisories, view broader audit
Knowledge Curator	validate extracted events and lessons
Admin	users, roles, integrations, configuration

61. Deployment Architecture

For the hackathon, the frontend can remain on the stable Vercel production project while the backend is hosted on a reachable service/tunnel. For the target OIL environment, frontend, FastAPI, PostgreSQL/PostGIS, Qdrant, Ollama and file storage should sit behind an internal reverse proxy inside company-controlled infrastructure.

- Demo: stable Vercel URL + temporary backend endpoint.
- Target: same-origin internal domain + TLS + secure cookies.
- Database: PostgreSQL 17 + PostGIS extension.
- Vector store: Qdrant.
- Inference: Ollama/Qwen locally; vLLM optional future scale-out.
- Streaming: WITSML/ETP adapter when real eRTMAC integration is available.

62. Observability and Reliability

- Health/readiness endpoints for API, DB, Qdrant and model gateway.
- Structured logs with request/execution IDs.
- Metrics: query latency, retrieval latency, risk evaluation latency, alert counts, false-positive feedback.
- Ingestion job state and failure reason.
- Telemetry freshness and lag.
- Deterministic retry/idempotency for durable workflows.
- Backup/restore for PostgreSQL and document store.
- Model digest/version pinning for repeatability.

63. API Plan

- Use explicit response schemas.
- Paginate lists.
- Return unavailable/null rather than fake zero.
- Include as_of/source metadata where relevant.

Endpoint family	Examples
Wells	GET /api/wells, /api/wells/{id}
Nearby	GET /api/wells/{id}/nearby?radius_km=...
Correlation	GET /api/wells/{id}/correlation
Events	GET /api/events?well_id=&formation=&type=
Risk	GET /api/wells/{id}/risk?lookahead_m=
Telemetry	GET /api/wells/{id}/telemetry?channels=&from=&to=
Knowledge	POST /api/query / evidence search
Advisories	GET/POST review/acknowledge
Ingestion	POST /api/ingest/report
Audit	GET /api/audit with bounded filters

64. Frontend Architecture

- Keep React 19 + Vite + existing router/session shell.
- Add MapLibre only to map/correlation views; lazy-load it.
- Use native chart library already selected or ECharts for depth/time series.
- Centralize API calls in services/api.js.
- Create domain components: WellSelector, RiskCard, OffsetTable, CorrelationTrack, EventChip, EvidenceDrawer.
- Keep public landing cinematic if desired; internal NWIS workbench should remain light, dense and operationally readable.
- Reduced-motion and keyboard accessibility must remain functional.

Key sources: R7

65. Migration from Sovereign Workbench

Do not delete old migrations, benchmark guards or security infrastructure during the overnight pivot. Create new additive NWIS modules and archive the old product branch.

Keep	Adapt	Hide/retire later
Auth, sessions, roles	Knowledge agent -> Well Knowledge Agent	P&ID viewer
FastAPI + PostgreSQL	Maintenance intelligence -> drilling-event intelligence	Refinery maintenance pages
Qdrant/RAG	Sensor pipeline -> drilling telemetry	Plant operations/startup/isolation
LangGraph + durable execution	Safety agent -> Drilling Risk Agent	MRPL branding/content
Audit + terms gate	Verified knowledge -> validated lessons	Old government-resource page if distracting
Ollama/Qwen gateway	Dashboard -> well/risk dashboard	Refinery datasets

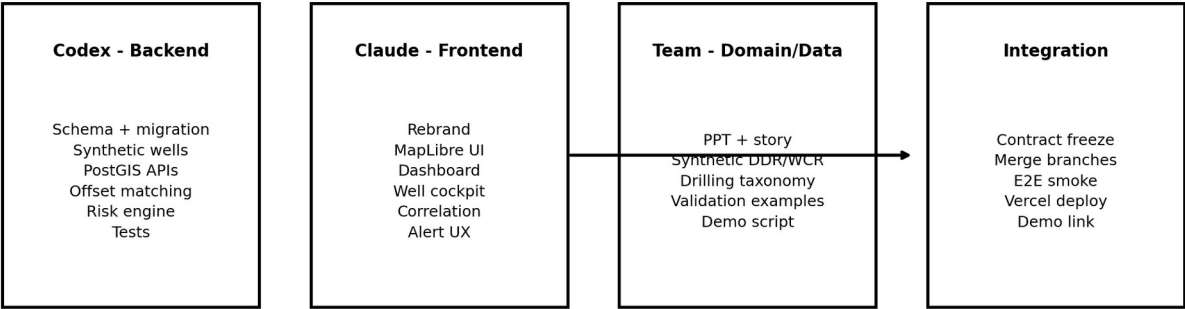
66. Repository Cleanup Plan

- Create archive/sih26117-sovereign-final branch and tag.
- Create pivot branches/worktrees for NWIS backend and frontend.
- Add docs/nwis/ as the new authoritative documentation root.
- Update README to point to NWIS docs and mark old phase docs historical.
- Create data/nwis_demo/ and keep refinery demo data outside the active seed path.
- Create benchmark/nwis/ rather than modifying frozen old benchmark files.
- After integration passes, remove dead navigation and then clean unreachable code in a separate commit.
- Never use git clean/reset during the pivot unless artifacts are explicitly reviewed.

67. Parallel Agent Strategy

Run Codex and Claude in separate worktrees. Codex owns backend/domain data. Claude owns frontend/product experience. Team members prepare domain content, synthetic reports and validation cases. Integration happens only after contracts are frozen.

Parallel Rebuild Workstreams



Rule: one writer per worktree; merge only after tests and API-contract review.

68. Overnight Sprint Plan

- Freeze scope after the first working vertical slice.
- Do not add Google OAuth, SMTP or complex live integrations unless core NWIS is already green.

Block	Target	Deliverable
0-30 min	Archive/rename/rebrand shell	stable PPT URL shows NWIS
30-120 min	Backend foundation	NWIS schema, seed, nearby API, events
30-120 min parallel	Frontend foundation	NWIS routes, dashboard, map, well page
120-210 min	Intelligence	offset score + basic risk + evidence
210-270 min	Integration	real API wiring + synthetic demo
270-330 min	E2E and fixes	happy-path demo green
330-390 min	Final deploy/PPT/video	stable link + screenshots

69. Team Task Breakdown

If only two active coding agents are available, combine A+B under Codex and C under Claude; the human team handles D.

Member	Primary ownership	Concrete tasks
Member A - Backend	data model + APIs	migrations, PostGIS, wells/events, radius endpoint
Member B - AI/Data	extraction + risk	synthetic corpus, event parser, similarity/risk engine, benchmark
Member C - Frontend	maps + cockpit	MapLibre, dashboard, well/offset/correlation/risk UI
Member D - Integration/Story	deployment + docs + demo	seed flows, E2E, PPT screenshots, demo script, references

70. Prioritized Backlog

Priority	Features
P0 - must demo	rebrand, well seed, map/radius, events, offset ranking, risk look-ahead, citations, deployed link
P1 - strong differentiators	formation correlation, telemetry replay, alert persistence, event validation
P2 - if time	WITSML adapter stub, OSDU mapping, voice query, advanced trajectory similarity
P3 - after SIH	real eRTMAC integration, production risk-model calibration, enterprise HA, full OSDU/WITSML connectors
Scope rule A polished P0 end-to-end demo beats ten incomplete “AI features”.	

71. Demo Flow for Judges

- 1. Landing: “eRTMAC sees the live well; NWIS remembers nearby wells.”
- 2. Login with demo account.
- 3. Dashboard: select active demo well.
- 4. Map: set radius and show nearby offsets.
- 5. Offset ranking: explain why two wells are more relevant than the closest well.
- 6. Correlation: align formation/depth and reveal historical stuck-pipe/loss events.
- 7. Risk: look ahead 100 m; show increased hazard with evidence.
- 8. Open “Why this alert?” and cite DDR/WCR passage.
- 9. Show live/simulated telemetry trend reinforcing or reducing risk.
- 10. Engineer acknowledges advisory; audit entry appears.
- 11. Close with architecture: local AI, PostGIS, Qdrant, WITSML-ready integration.

Demo timing

Aim for 2.5-3.5 minutes. Avoid waiting for long model generation; pre-seed evidence and keep one deterministic scenario.

72. PPT Mapping

- Do not claim measured NPT reduction without data.
- Use “designed for OIL/eRTMAC workflows”, not “deployed in OIL”.
- Prototype screenshots should show map + risk look-ahead, not old refinery pages.

Slide	Recommended NWIS content
1	SIH26121 title, Oil India Limited, team, stable prototype URL
2	Problem: live data without institutional memory -> proposed NWIS workflow
3	Technical approach: documents + eRTMAC -> structuring -> PostGIS/Qdrant -> correlation -> risk -> advisory
4	Feasibility: reuse local AI/RAG + public/synthetic data + WITSML readiness; challenges/mitigations
5	Impact: faster offset knowledge, proactive risk visibility, traceability; real prototype screenshots
6	Research: WITSML/ETP, OSDU, PostGIS, public datasets, drilling-risk literature

73. Key Risks and Mitigations

Risk	Mitigation
No official dataset	Synthetic labelled demo + public validation datasets; clearly disclose origin
Domain mismatch from old project	new canonical well/event schemas; hide refinery UI
False confidence from AI	confidence decomposition + evidence + human review
Rare event imbalance	hybrid/rules/anomaly approach + PR-AUC/calibration
Bad OCR numbers	numeric validation + source spans + review queue
Geology overgeneralization	formation/depth aware matching; show uncertainty
Alert fatigue	persistence/hysteresis/cooldown
Time pressure	vertical-slice P0 scope + parallel worktrees
Deployment link instability	reuse one Vercel production project and stable URL

Key sources: R12, R13

74. Feasibility and Business Value

Technical feasibility is high because the difficult platform foundations already exist in the previous workbench: local inference, RAG, document processing, PostgreSQL/Qdrant, authentication, audit and React/FastAPI. The new engineering effort is primarily domain modeling, spatial search, correlation, risk scoring and frontend conversion.

Operational value should be framed around reduced knowledge-retrieval time, better reuse of historical lessons, earlier visibility of offset-well hazards, consistent evidence access and less dependence on individual memory.

Quantitative NPT savings should be treated as a future validation KPI, not a guaranteed prototype result.

- Scalable to field-level well inventory through indexed spatial queries.
- Supports structured and unstructured data.
- Can operate as a standalone decision-support layer alongside eRTMAC.
- Interoperability path through WITSML/ETP and OSDU concepts.
- Human-centered governance reduces risk of automation overreach.

Key sources: R1, R2, R3

75. Research and Technology References

The following references are the main external foundations used in this report. They should also inform the final SIH PPT and technical Q&A.

- R1 - SIH26121 problem statement mirror: eRTMAC-NWIS, Oil India Limited, SIH 2026.
- R2 - Oil India Limited, Technology and Innovation / Drilling pages describing eRTMAC real-time monitoring.
- R3 - Oil India eRTMAC tender CDH6605P21 / later tenders describing WITS/WITSML, real-time and historical analytics.
- R4 - Energistics WITSML 2.1 and ETP 1.2 developer specifications.
- R5 - OSDU Wellbore Domain Services.
- R6 - PostGIS ST_DWithin and spatial-index guidance.
- R7 - MapLibre GL JS documentation.
- R8 - Equinor Volve public data release.
- R9/R10 - Qdrant hybrid search / RRF documentation.
- R11 - Equinor FORCE 2020 well logs.
- R12/R13/R14 - drilling stuck-pipe/risk-prediction research demonstrating value of time-series indicators and ML.
- R15 - BAAI BGE embeddings/reranker documentation.

Research discipline

Use external literature to justify architecture and evaluation methods; do not copy unverified claimed savings or performance from other SIH teams.

76. Glossary for Teammates

Term	Meaning
eRTMAC	Oil India enhanced real-time monitoring and analytics centre for drilling
NWIS	Nearby Wells Intelligence System
Offset well	historical/nearby well used as an analog
WCR	Well Completion Report
DDR	Daily Drilling Report
MD	Measured Depth along the well path
TVD	True Vertical Depth
TVDSS	TVD relative to sea-level/subsea datum
WITSML	industry data standard for well/drilling data
ETP	Energistics Transfer Protocol
OSDU	open subsurface data platform ecosystem
NPT	Non-Productive Time
ROP	Rate of Penetration
WOB	Weight on Bit
RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
HITL	Human in the Loop

Key sources: R4, R5

77. Appendix A - Example Risk API Payload

A compact contract keeps frontend and backend teams synchronized. The following is illustrative, not final production schema.

- Probability and confidence are different fields.
- The response names evidence IDs and offset wells.
- Missing values should be null/unavailable, not fabricated.

GET /api/wells/{id}/risk?lookahead_m=100

```
{
  "well_id": "ACTIVE-01",
  "as_of": "2026-09-29T16:00:00Z",
  "current_md_m": 2450.0,
  "formation": "TIPAM_A",
  "lookahead_m": 100,
  "hazards": [
    {"type": "stuck_pipe", "probability": 0.72, "confidence": 0.78,
      "trend": "rising", "supporting_offset_wells": ["OFF-04", "OFF-09"],
      "top_factors": ["formation_match", "historical_event_density", "torque_trend"],
      "evidence_ids": ["EVT-104", "CHUNK-982"]}
  ]
}
```

78. Appendix B - Build Checklist / Definition of Done

- Repository archived and pivot branch/worktrees created.
- Stable Vercel production URL shows NWIS branding.
- Synthetic dataset seed is reproducible.
- At least 10 wells visible on map.
- Radius search works via backend.
- At least 3 event types extracted/stored with evidence.
- Offset ranking exposes component scores.
- Formation/depth correlation renders.
- One risk look-ahead scenario works deterministically.
- One alert links to at least two source/evidence records.
- Demo account can access core screens without admin rights.
- Lint/tests/build pass.
- Backend targeted tests pass.
- No old MRPL/refinery wording on visible NWIS routes.
- PPT screenshots come from current NWIS build.
- Demo script practiced under 4 minutes.
- README and docs/nwis explain how teammates run the system.

Final acceptance

The project is demo-ready when a teammate who did not write the code can start the stack, log in, follow the judge flow and explain where every risk statement came from.

Key sources: R1